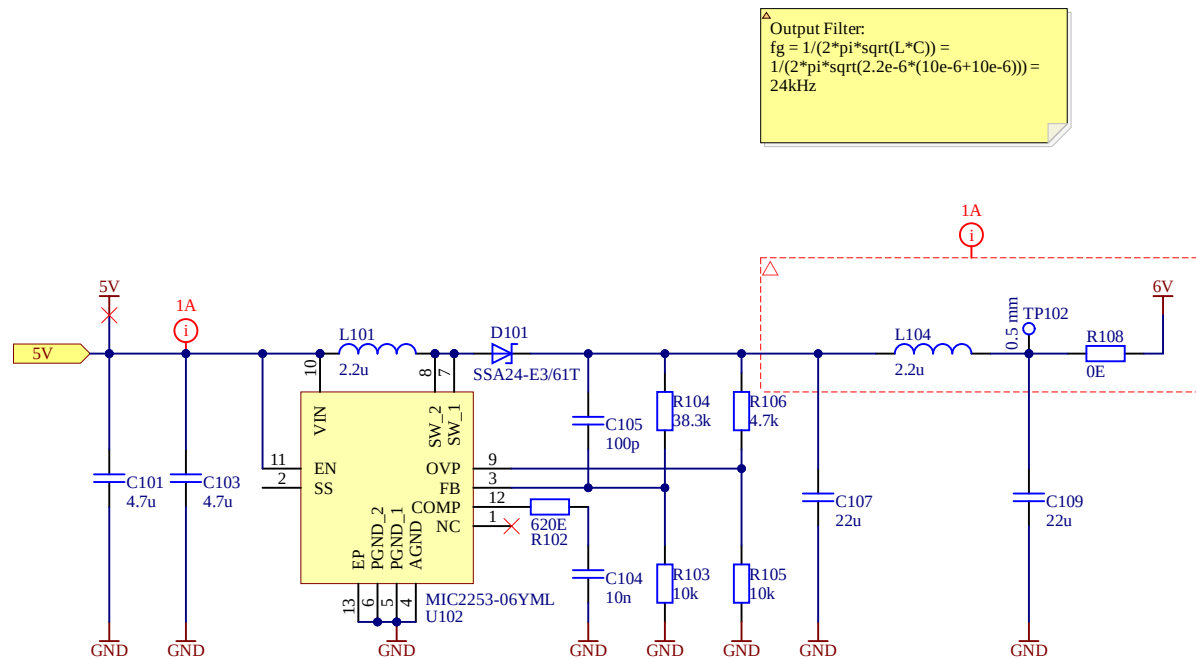
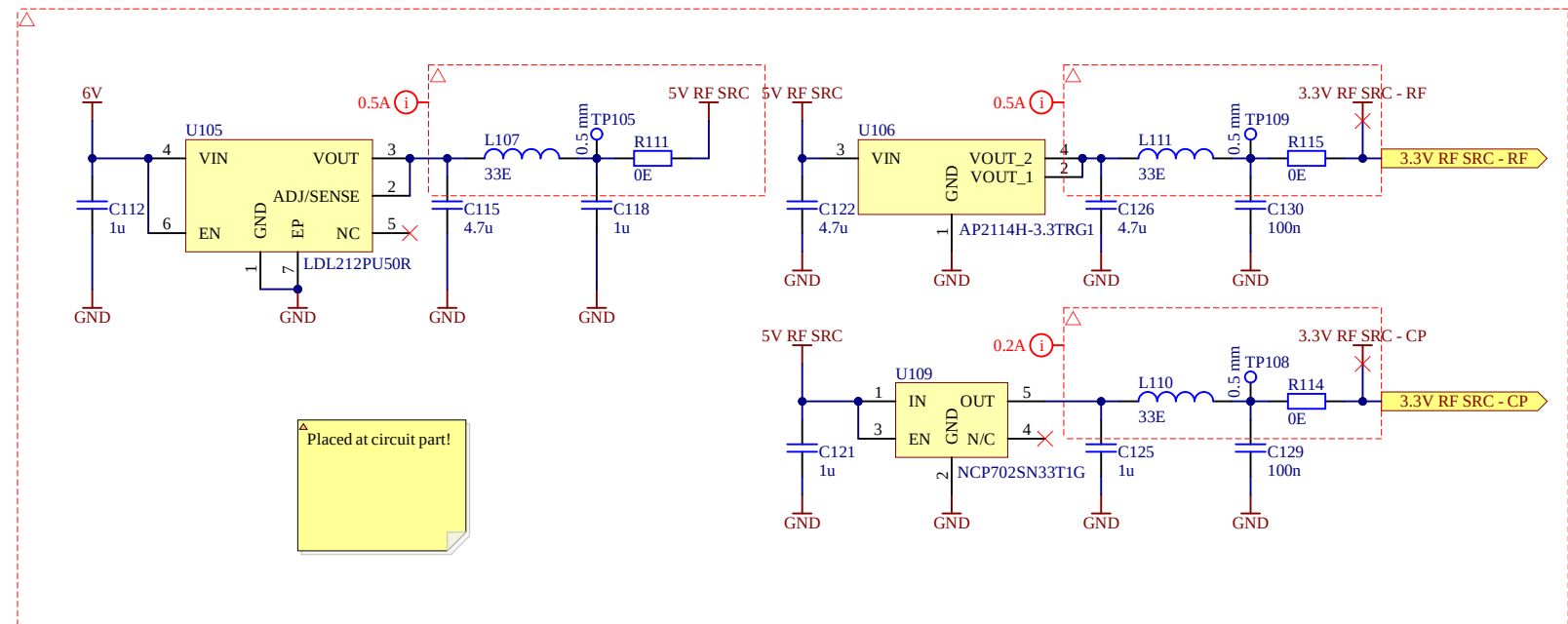




Output Filter:

$$f_g = \frac{1}{2\pi \sqrt{L \cdot C}} = \frac{1}{2\pi \sqrt{2.2 \times 10^{-6} (10 \times 10^{-6} + 10 \times 10^{-6})}} = 24 \text{ kHz}$$



Placed at circuit part!

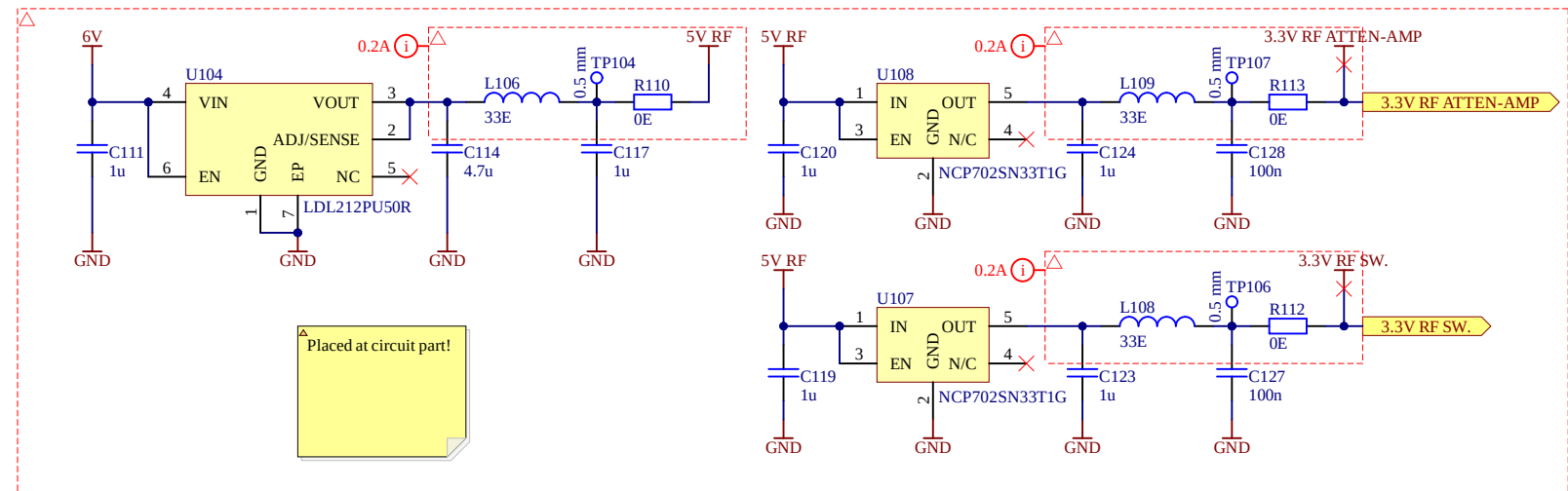
$L = 2.2\mu\text{H}$ as per reference design
 $C_{out} = 22\mu$ as per reference design

 Set ss to 10ms
 $C_{ss} = (T_{ss} - 1\text{ms}) / 85k\Omega = (10\text{ms} - 1\text{ms}) / 85k\Omega = 105\text{nF} \rightarrow 100\text{nF}$ (10nF internal cap!)

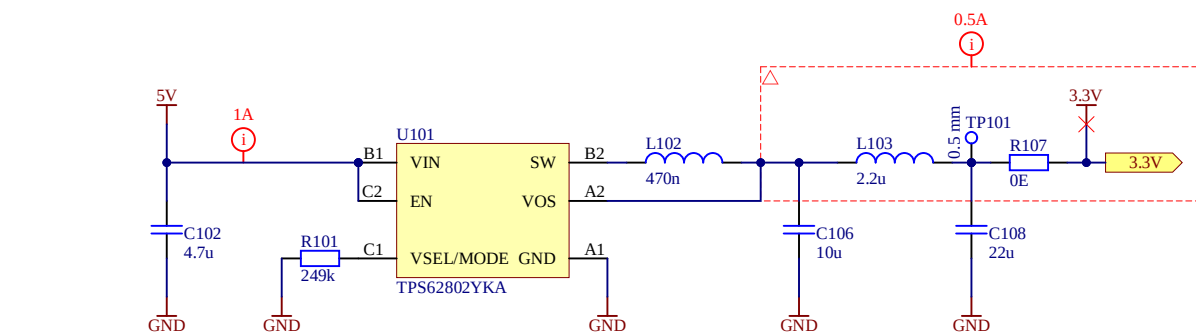
 $U_{ovp} = 8\text{V}$
 $R_{bottom} = 10k$
 $R_{top} = (U_{ovp} \cdot 15k\Omega \cdot R_{bottom}) / (1.245 \cdot 67k\Omega) - R_{bottom} = (8\text{V} \cdot 15k\Omega \cdot 10k\Omega) / (1.245 \cdot 67k\Omega) - 10k\Omega = 4.39k\Omega \rightarrow 4.7k\Omega$

 $U_{out} = 6\text{V}$
 $U_{fb} = 1.245\text{V}$
 $I_{fb} = -450\text{nA}$
 $R_{bottom} = 10k\Omega$
 $R_{top} = (U_{out} - U_{fb}) / (U_{fb} / R_{bottom} + I_{fb}) = (6\text{V} - 1.245\text{V}) / (1.245\text{V} / 10k\Omega - 450\text{nA}) = 38.3k\Omega$

 $C_{comp} = 10\text{nF}$ as per reference design
 $R_{comp} = 620\Omega$ as per reference design

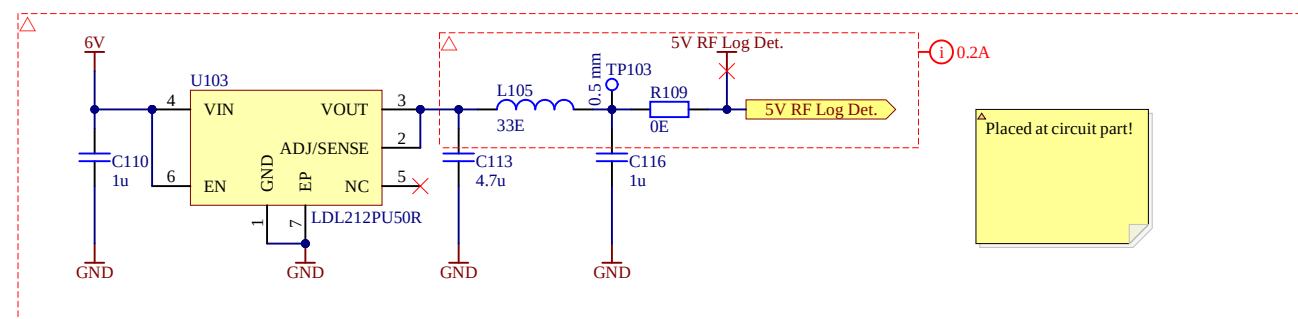


Placed at circuit part!



Design from WEBENCH

Output Filter:
 $f_g = 1 / (2 * \pi * \sqrt{L * C}) =$
 $1 / (2 * \pi * \sqrt{2.2e-6 * (10e-6 + 10e-6)}) =$
 24kHz

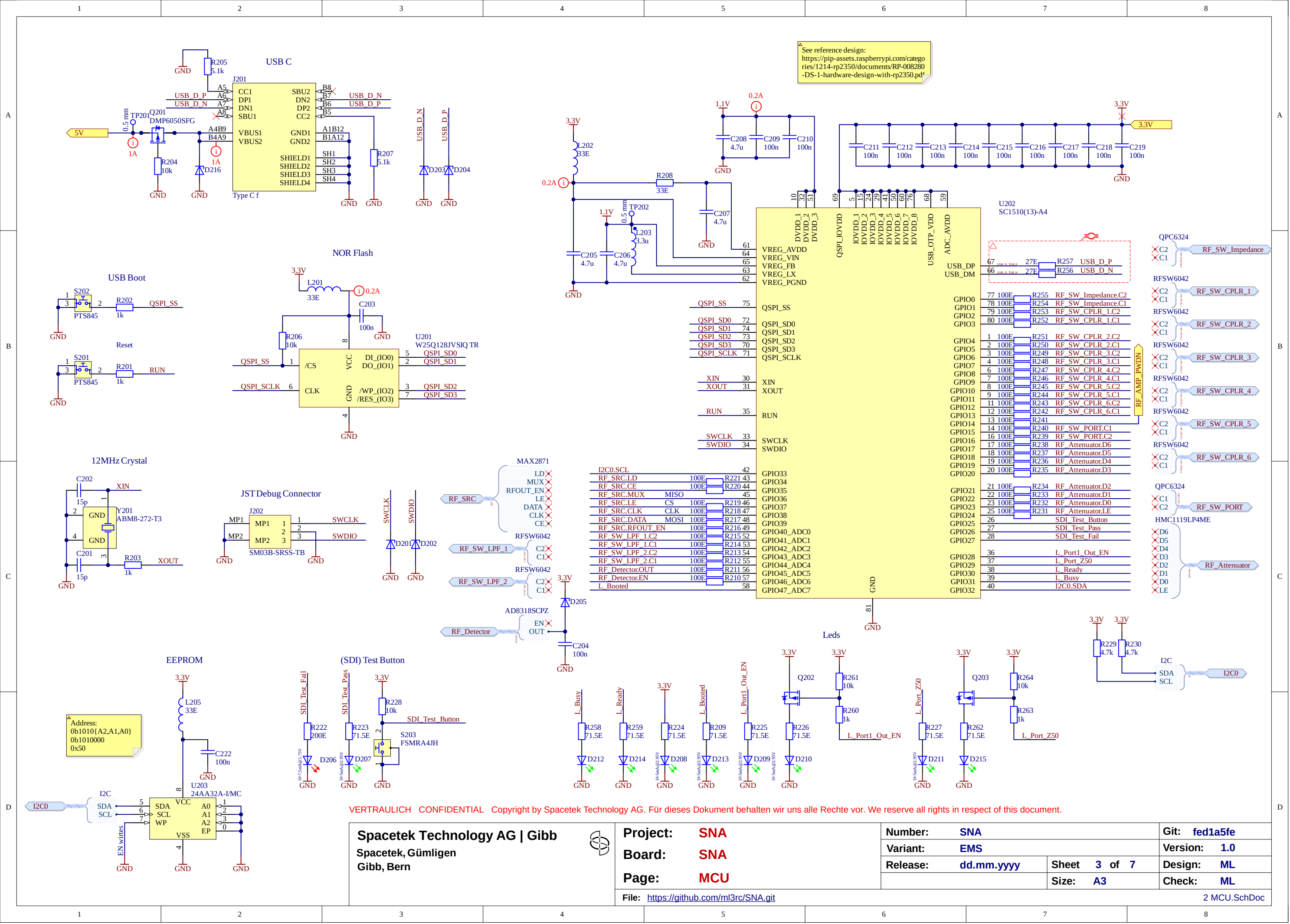


Placed at circuit part!

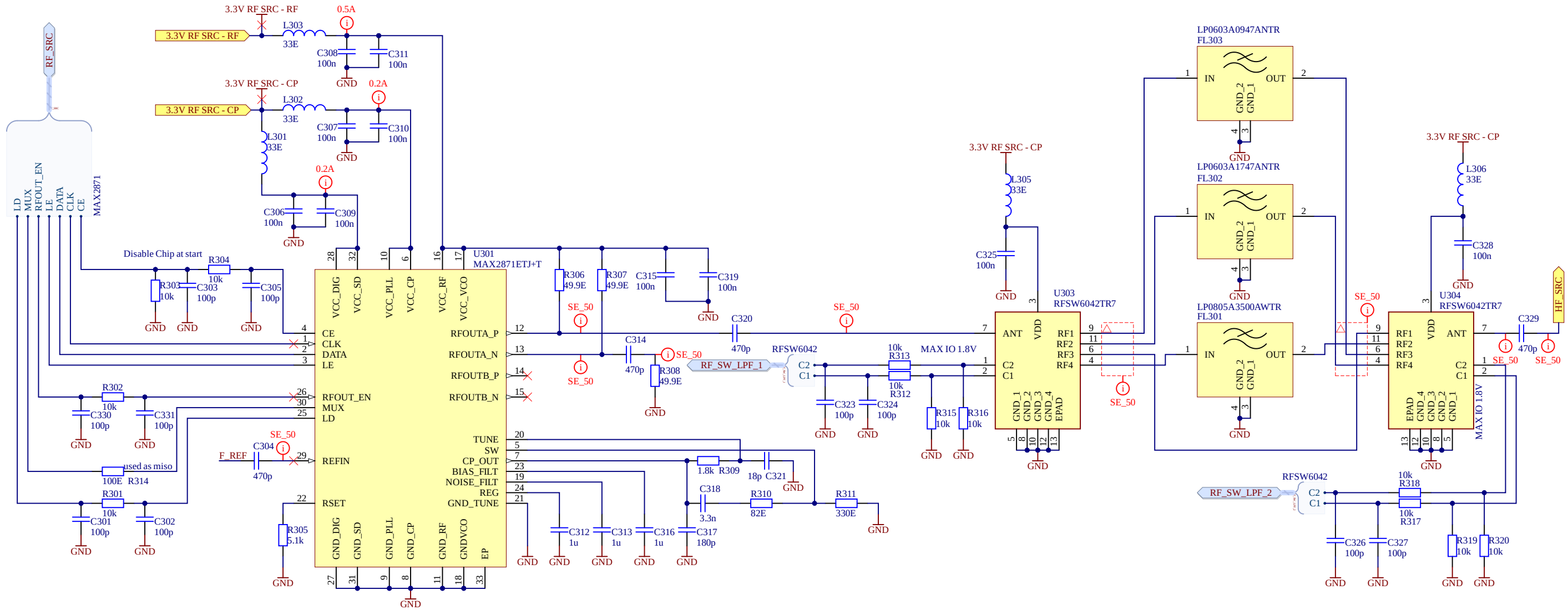
For current specs see power blockdiagramm
 20mil -> 1A (2.14A)
 15mil -> 0.5A (1.79A)
 12mil -> 0.2A (1.56A)

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See LibreVNA Schematics for reference



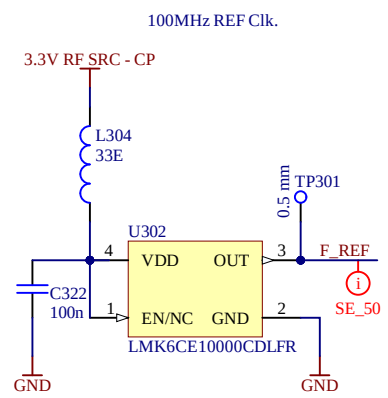
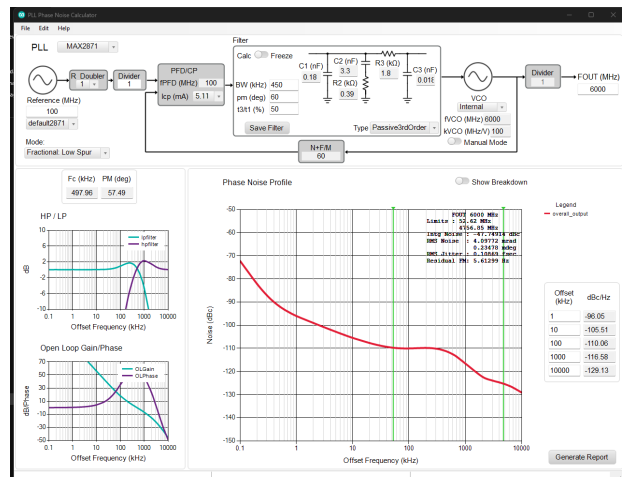
Rset = 5.1kOhm as per datasheet

C_BIAS_FILT, C_NOISE_FILT and C_REG = 1uF as per datasheet

The loop filter can be calculated with:
<https://www.analog.com/en/resources/evaluation-hardware-and-software/embedded-development-software/software-download.html?swpart=SFW0007420A>

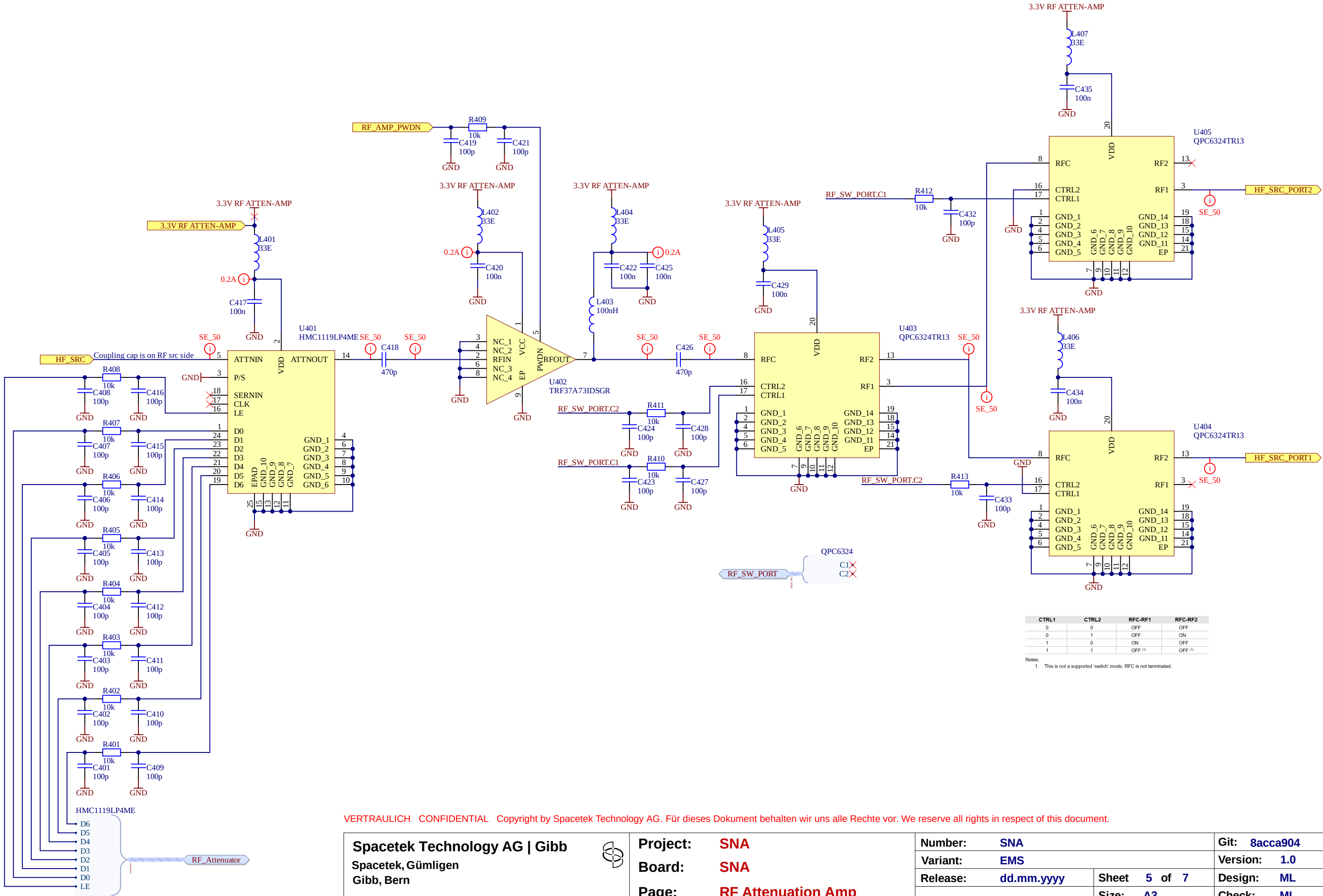
f_bandwidth_loopfilter = ~450kHz -> fast response, but higher noise

$R_{Cr_coupling} = ((1)/(2*\pi*50MHz*470pF)) = 6.8Ohm$



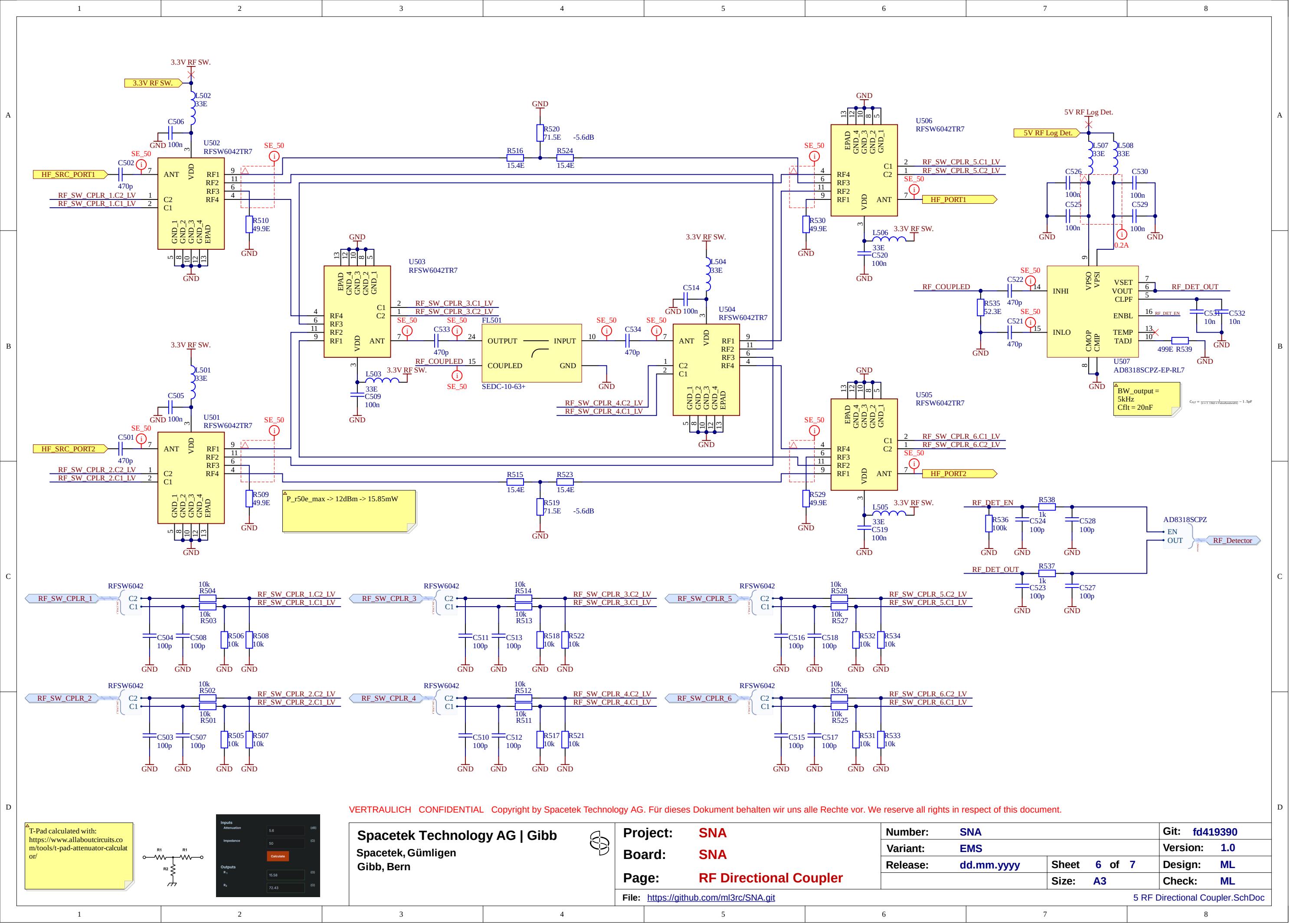
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Spacetek Technology AG Gibb Spacetek, Gümligen Gibb, Bern		Project:	SNA	Number:		SNA		Git:	8acca904		
		Board:	SNA	Variant:		EMS		Version:	1.0		
		Page:	RF Attenuation Amp	Release:	dd.mm.yyyy	Sheet	5	of	7	Design:	ML
						Size:	A3	Check:	ML		
				File: https://github.com/ml3rc/SNA.git							



50-75 Matching Pad:
 $\text{solve}(\{75=((1)/(((1)/(R_p))+((1)/(50))))+R_s, 50=((1)/(((1)/(75+R_s))+((1)/(R_p))))\})$
 $R_p = 86.6\Omega$
 $R_s = 43.3\Omega$
Insertion Loss = 5.72dB

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Project: **SNA**
Board: **SNA**
Page: **RF Ports**

File: <https://github.com/ml3rc/SNA.git>

Number: **SNA**

Variant: **EMS**

Release: **dd.mm.yyyy**

Sheet **7 of 7**

Size: **A3**

Git: **fed1a5fe**

Version: **1.0**

Design: **ML**

Check: **ML**

6 RF Ports.SchDoc